Dynamic Spatiotemporal Patterns of Nighttime Fishing Vessel Activity Across Indonesia's Fisheries Management Zones

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Abstract

Satellite-based monitoring is increasingly critical to global fisheries governance in archipelagic, tropical waters where small-scale fleets evade conventional tracking, and Indonesia's extensive, densely fished seas make it a demanding test case for such approaches. This study aims to characterize the spatio-temporal dynamics of nighttime fishing vessel activity across the 11 Fisheries Management Areas of the Republic of Indonesia (WPPNRI) using Visible Infrared Imaging Radiometer Suite Day/Night Band (VIIRS DNB) data over the 2012 to 2026 observation period, and to evaluate forecasting performance in regions with contrasting activity density. An integrated quantitative approach was applied, comprising radiance classification, Mann-Kendall trend analysis and Sen's Slope estimation, Seasonal-Trend Decomposition using LOESS (STL), and a comparison of Error, Trend, Seasonal (ETS) and Seasonal Autoregressive Integrated Moving Average (SARIMA) models. The results show that detected activity was dominated by the medium (53.7%) and low (41.8%) radiance classes, with the highest spatial concentration in WPP 712. Trend analysis revealed a spatial reorganization characterized by stagnation in several western regions and increasing activity in several eastern WPPs, forming an Eastern Frontier Expansion pattern. Overall activity exhibited a strong seasonal pattern, peaking in the third quarter, coinciding with the Southeast Monsoon. In the forecasting evaluation for WPP 712 and WPP 717, ETS produced lower descriptive error than SARIMA, including a MAPE of 17.5% and RMSE of 264.2 for WPP 717, although the difference was not statistically significant based on the Diebold-Mariano test. These findings indicate the potential of VIIRS as a complementary

source of spatio-temporal information to support region-based fisheries monitoring and management.

Keywords:
VIIRS DNB; nighttime fishing activity; spatio-temporal; Indonesian Fisheries Management Areas; forecasting.

1. Introduction

As a maritime nation with a vital ecological and economic role, Indonesia's marine capture fisheries sector contributes substantially to the global supply of animal protein and to food security (Phan et al., 2024). Sustainable governance of national maritime fisheries is crucial to supporting Sustainable Development Goal 14 (Life Below Water), which calls for the optimal management of marine resources to curb overexploitation (Phan et al., 2024; Pauly & Zeller, 2016). One of the main pillars of the capture fisheries sector in Indonesian waters is light-attraction fishing, which uses high-intensity light to aggregate small pelagic species and squid (Hsu et al., 2019; Ruiz et al., 2019; Seto et al., 2023). Nevertheless, fishing fleets frequently operate without activating the Automatic Identification System (AIS), creating substantial spatial data gaps (Kroodsma et al., 2018; Seto et al., 2023). The absence of tracking transmissions gives rise to the phenomenon of dark vessels, which carry a high risk of enabling unregulated fishing activity (IUU fishing) (Park et al., 2020). In this context, satellite remote sensing instruments, particularly the VIIRS DNB sensor (Elvidge et al., 2015), represent a major technological breakthrough. The continually improving capabilities of the VIIRS DNB sensor have proven able to detect the light emissions of dark fleets precisely and consistently, surpassing the limitations of conventional monitoring and AIS/VMS systems in Indonesian Fisheries Management Areas (Zuo et al., 2024; Hsu et al., 2019).

Although the use of VIIRS DNB-based remote sensing has opened opportunities to expand the monitoring of nighttime fishing activity, understanding of how this activity changes spatially and temporally across Indonesia's various waters remains limited. Vessel tracking-based monitoring has also not yet been able to provide a comprehensive picture of fleet distribution, so satellite nighttime light detection can serve as a complementary source of information for identifying vessel activity that is difficult to observe through conventional systems (Hsu et al., 2019; van Beek et al., 2021). This issue is increasingly relevant because fishing pressure and activity vary in character across regions. The Java Sea, for example, has long faced high fishing capacity pressure, whereas the Arafura Sea requires evidence-based management and resource-use regulation to maintain fisheries sustainability (Squires et al., 2003; Afriansyah et al., 2024; Wudianto et al., 2024). Fishing activity also does not occur at

a constant level throughout the year, but is influenced by seasonal dynamics and oceanographic conditions, including monsoon variability and upwelling in Indonesian waters (Wirasatriya et al., 2021). VIIRS-based studies have likewise demonstrated the ability of nighttime light data to identify the distribution and temporal variation of vessel activity, but most research has focused on specific regions and has not yet provided an integrated picture of long-term activity change across WPPs (Hsu et al., 2019; Li et al., 2023). In addition, the interpretation of vessel light detections presents technical challenges, including the effect of moonlight, which can influence detection consistency (Yoon et al., 2026). These conditions highlight the need for analyses capable of identifying spatial distribution, long-term trends, seasonal patterns, and inter-regional differences in the temporal characteristics of fishing activity.

A number of earlier studies have examined lit-fishing activity using VIIRS DNB satellite data. Research conducted at both global and regional scales has successfully mapped dark vessels, evaluated the effectiveness of fishery closures, and highlighted the challenges of high-seas governance in regions such as the North Pacific and the South China Sea (Park et al., 2020; Elvidge et al., 2020; Li et al., 2024; Wang et al., 2025). However, these studies have generally focused on industrial fleets and overlooked the unique dynamics of medium-scale to traditional fishing fleets, which in fact dominate tropical archipelagic waters. Meanwhile, local studies in Indonesia are generally still limited to very short time spans and focused on single regions, and are more often aimed simply at cross-validation with Vessel Monitoring System (VMS) tracks without integrating trend analysis holistically (Hsu et al., 2019; Van Beek et al., 2021; Nurholis et al.). In terms of integrated monitoring and spatial resolution, although several VIIRS satellite-based ecological reporting systems have been initiated over the past decade (Rogers et al., 2023; Tian et al., 2022; Tian et al., 2024), this literature has not yet explored the comparison of predictive algorithms across contrasting-density data classifications. The novelty and principal contribution of this study lie in an integrated analytical approach comprising: (1) an evaluation of the historical time series at the full regional scale over a 14-year period (2012 to 2025) across all 11 WPPNRI based on light radiance intensity; (2) a comprehensive decomposition of monsoon seasonal variables using the Seasonal-Trend Decomposition using LOESS (STL) method; (3) identification of the spatial reorganization of fishing activity through the discovery of the Eastern Frontier Expansion pattern; and (4) a head-to-head reliability test of the ETS forecasting model against the linear SARIMA model in waters with extreme activity density characteristics (highest and lowest).

Although recent research has demonstrated significant progress in the use of VIIRS to monitor nighttime fishing, essential gaps remain in the integration of cross-regional spatial scale, long-term temporal dimensions, and forecasting. Specific studies such as those conducted by Li et al. (2023) in the Bohai Sea, De Ocampo (2024) in Philippine waters, and

Ahn and Yang (2025) in Korean jurisdictional waters have indeed succeeded in mapping vessel distribution spatio-temporally, but these analyses have not been directed toward multi-decade decomposition trend modeling or cross-regional forecast comparison. On the other hand, much of the recent literature instead concentrates on resolving technical problems, namely improving the accuracy of vessel pixel extraction or detection through machine learning and deep learning architectures, such as the application of CNN, YOLO, and increasingly complex attention network models (Miao et al., 2022; Song et al., 2023; Tsuda et al., 2023; Zuo et al., 2024; Yoon et al., 2026). As a result, the main contribution of these studies centers more on optimizing remote sensing software engineering rather than on providing policy-relevant outputs related to shifts in fishing activity. In the Indonesian management context, the use of VIIRS remains far behind in macro-level analytics. Current research is largely confined to isolated spatial distribution observations, such as in the Aru Islands and West Sumatra waters (Van Beek et al., 2021; Nurholis et al.; Hsu et al., 2019). This condition leaves a sharp research gap, given the absence of a spatio-temporal and econometric evaluation framework across WPPs that can not only unravel the phenomena of stagnation and activity expansion at the full regional scale, but also provide a predictive forecasting foundation that is essential for future resource allocation planning in Indonesian fisheries management.

This study therefore aims to characterize the spatio-temporal dynamics of nighttime fishing vessel activity across the 11 WPPNRI during the 2012 to 2026 period using VIIRS DNB data, and to evaluate forecasting performance in regions with differing activity density characteristics. This objective is grounded in the capability of VIIRS to detect vessel activity through nighttime light emissions and its potential as a complementary source of information for conventional vessel monitoring systems (Hsu et al., 2019; van Beek et al., 2021). Specifically, this study aims to: (1) map the spatial distribution and identify the activity density characteristics of nighttime fishing vessels across the 11 WPPNRI based on VIIRS light detection, consistent with previous research showing that VIIRS data can be used to identify the spatial distribution and concentration of fishing vessels (Li et al., 2023; Sarangi & Jaiganesh, 2022); (2) analyze long-term trends and changes in the distribution of fishing activity across regions during 2012 to 2025 to identify changes in intensity and the reorganization of fishing activity; (3) identify the trend and seasonal components of fishing activity and their relationship to the monsoon cycle, given that VIIRS-based research has shown that vessel activity can exhibit strong temporal and seasonal variation (Li et al., 2023; Sarangi & Jaiganesh, 2022); and (4) compare the performance of the ETS and SARIMA models in forecasting fishing activity in the regions with the highest and lowest activity to evaluate predictive capability under differing activity density characteristics. The findings of this study are expected to provide an empirical picture of the distribution, trends, spatial reorganization, and seasonal patterns of nighttime fishing activity, as well as to provide predictive information that supports spatially based fisheries monitoring and management.

2. Methodology

This study employed a quantitative observational design based on remote sensing to identify the spatio-temporal patterns of light-assisted nighttime fishing activity in Indonesian waters. The primary data consisted of detection events from the VIIRS Boat Detection (VBD) product, which were subsequently integrated with the polygon boundaries of the 11 WPP RI. The analytical framework comprised data collection, pre-processing, and seven statistical analysis components. All stages were directed toward explaining the location of detection concentrations, the distribution of radiance intensity, differences in profile across WPP RI, long-term trends, interannual variability, seasonal patterns, and the forecasting capability for light-assisted nighttime fishing activity.

The unit of observation in this study is a single VBD record at a specific time and coordinate, rather than the unique identity of an individual vessel or the volume of catch. In this study, fishing activity is operationally defined as the frequency of vessel light detection events at sea which, after quality filtering and spatio-temporal aggregation, are used as a proxy for the relative pattern of light-assisted nighttime fishing activity. The empirical basis for this definition comes from the matching of VBD with VMS data in Indonesia, which showed that 96.42% of successfully matched detections occurred while a vessel was engaged in fishing (Hsu et al., 2019). This evidence applies to the subset of detections successfully matched with VMS; therefore, each individual VBD record is not treated as definitive confirmation of a fishing event.

2.1. Data Collection

2.1.1. VIIRS Boat Detection Data

The primary data for this study were obtained from the VBD product developed by the Earth Observation Group (EOG), Payne Institute for Public Policy, Colorado School of Mines, based on Day/Night Band (DNB) observations from the VIIRS instrument carried by the Suomi National Polar-orbiting Partnership (Suomi NPP) satellite. The DNB sensor is capable of recording low-intensity visible light sources at night with a nominal spatial resolution of approximately 742 m, allowing it to detect point light sources at sea, including vessels using lamps to attract fish. A number of studies have shown that VIIRS and nighttime remote sensing data can represent the spatio-temporal dynamics of light-assisted fisheries, particularly for fleets with relatively high light intensity (Exeter et al., 2021; Li et al., 2021; Tian et al., 2022). The VBD archive used covers the period from April 2012 to July 2026 and was obtained in Comma-Separated Values (CSV) format, comprising 5,041 files totaling 20.03 GB and 18,115,494 raw records. Each record represents a single detection event at a specific time and location, including coordinates, observation time, radiance value

(Rad_DNB), and detection quality attributes. The Rad_DNB value is expressed in $nW\ cm^{-2}\ sr^{-1}$ and represents the light radiance received by the sensor; this value does not directly represent vessel size, tonnage, engine capacity, gear type, fishing effort, or catch outcome.

To ensure data quality, this study retained only records with *QF_Detect = 1*, the boat/strong detection category with the highest detection confidence level in the VBD product (Elvidge et al., 2015; Hsu et al., 2019). The relationship between vessel light detection and light fishing activity has been supported by research based on matching VBD with VMS, vessel position data, and fishing operation records. Hsu et al. (2019), for example, showed a high VIIRS detection rate for vessels identified as actively fishing, while Li et al. (2021) demonstrated agreement between nighttime satellite observations and light fishing operations in the South China Sea. Other studies have also shown that nighttime detection patterns can reflect the distribution and dynamics of fishing activity once non-fishing light sources, such as ports, docks, and shipping lanes, are accounted for (Exeter et al., 2021; Tian et al., 2022). Accordingly, after quality filtering, the frequency of VBD events was aggregated across spatial and temporal dimensions and used as a relative proxy for light-assisted nighttime fishing activity. This proxy is used to compare the concentration and change of activity across regions and periods, rather than as a direct measurement of vessel numbers, fishing-effort intensity, gear type, or catch volume. With this approach, the VBD data were subsequently used to construct the spatial distribution, identify activity concentrations, analyze temporal change, and evaluate fishing activity patterns across the WPPNRI.

2.1.2. Fisheries Management Area Boundaries

The distribution of the eleven WPPNRI that constitute the study area is shown in Figure 1. The map deliberately shows only the regional boundaries and does not yet include VBD points, because its function is to define the spatial framework of the analysis. During pre-processing, each VBD detection coordinate was assigned to a single WPPNRI through a point-in-polygon operation. Statistics for each WPPNRI were therefore calculated from detection events that genuinely fall within the corresponding polygon, rather than from proximity of points to a label or regional centroid.



Figure 1. Spatial coverage of the study area across the 11 Fisheries Management Areas of the Republic of Indonesia (WPPNRI).

The spatial analysis boundaries used the official WPPNRI vector data from the Directorate General of Capture Fisheries, Ministry of Marine Affairs and Fisheries of the Republic of Indonesia. This dataset comprises 11 management polygons, namely WPPNRI 571, 572, 573, 711, 712, 713, 714, 715, 716, 717, and 718. The WPP polygons were used as the analytical boundary, not merely as a bounding box. The bounding boxes in Table 1 only describe the general geographic range of each region, whereas the WPP code assigned to each detection was determined through a point-in-polygon operation against the actual polygon geometry.

Table 1. Geographic Boundaries and Coordinate Ranges of the 11 WPPNRI Regions

WPP RI	Water Area Coverage	Geographic Coordinate Range (Bounding Box)
571	Malacca Strait and Andaman Sea	94,49°-103,42° BT; 0,69°-8,21° LU
572	Indian Ocean West of Sumatra and Sunda Strait	91,64°-106,04° BT; 8,87° LS-6,42° LU
573	Indian Ocean South of Java to Nusa Tenggara, Sawu Sea, and Western Timor Sea	102,58°-126,50° BT; 13,94°-6,80° LS
711	Karimata Strait, Natuna Sea, and North Natuna Sea	102,90°-111,05° BT; 4,16° LS-7,74° LU
712	Java Sea	105,81°-116,27° BT; 7,94°-2,85° LS
713	Makassar Strait, Bone Bay, Flores Sea, and Bali Sea	114,42°-122,68° BT; 8,74° LS-1,02° LU
714	Tolo Bay and Banda Sea	121,29°-133,12° BT; 8,46°-0,83° LS
715	Tomini Bay, Maluku Sea, Halmahera Sea, Seram Sea, and Berau Bay	120,00°-134,97° BT; 5,29° LS-2,59° LU
716	Sulawesi Sea and Waters North of Halmahera Island	117,00°-131,63° BT; 0,74°-6,67° LU
717	Cenderawasih Bay and Pacific Ocean	128,69°-141,40° BT; 3,40° LS-4,27° LU
718	Aru Sea, Arafura Sea, and Eastern Timor Sea	127,42°-141,02° BT; 10,83°-4,25° LS

The VBD coordinates and WPP boundaries were first standardized to the WGS 84 geographic coordinate system (EPSG:4326). This standardization ensured that every detection point and WPP polygon shared the same spatial reference before the clipping and spatial join procedures were carried out in QGIS. For analyses that depend on distance and area, such as Kernel Density Estimation (KDE), the data were reprojected to a metric coordinate system so that the search radius and grid cell size would not be calculated in degree units.

2.1.3. Temporal Coverage and Data Interpretation Limits

The observation period comprises the annual series from 2012 to 2025 and one partial period extending to July 2026. Observation completeness was checked on a yearly basis according to the available file archives. All data available through July 2026 can be used to describe the cumulative spatial distribution and radiance distribution.

The likelihood that an activity is recorded as a VBD detection is influenced by lamp use and configuration, cloud cover, moon illumination, viewing angle, the position of the light source within the pixel, and the inherent sensitivity limitations of the sensor. Vessels that do not use strong lighting, such as some longliners or trawlers, may go undetected, while the deck lighting of non-fishing vessels may occasionally be recorded. This study should therefore be interpreted as reflecting the relative pattern of light-assisted nighttime fishing activity at the aggregate WPPNRI and time level, rather than as a complete census of fishing vessels or evidence of catch outcomes.

2.2. Computational Model and Statistical Analysis

The computational framework was built as a sequential pipeline from raw data to interpretable results. The first stage involved the collection and organization of VBD data and the WPPNRI boundaries. The second stage comprised cleaning, missing-value handling, quality filtering, spatial join, temporal aggregation, and analytical feature construction. The third stage was statistical modeling through seven pathways: spatial density analysis, radiance distribution, WPPNRI and radiance-class profiling, long-term trend analysis, interannual variability, seasonal pattern analysis, and forecasting and model performance evaluation. The relationship among these three stages is presented in Figure 2.

This modular workflow explicitly separates the geospatial data preparation stages (Stages 1 to 2) from the statistical analytical modeling stage (Stage 3). This approach ensures that the spatial filtering and temporal feature reconstruction processes are fully completed before the data enter the analytical chain. By centralizing the cleaned dataset (Final Boat Data) into a single Statistical Analysis node, numerical consistency and geospatial reference accuracy can be guaranteed across all modeling branches. This structural separation also minimizes the risk of computational redundancy, improves script execution efficiency, and

preserves the reproducibility of the final results, so that every map visualization, trend chart, seasonal diagram, and forecasting model parameter can be traced back precisely to its source raw data.

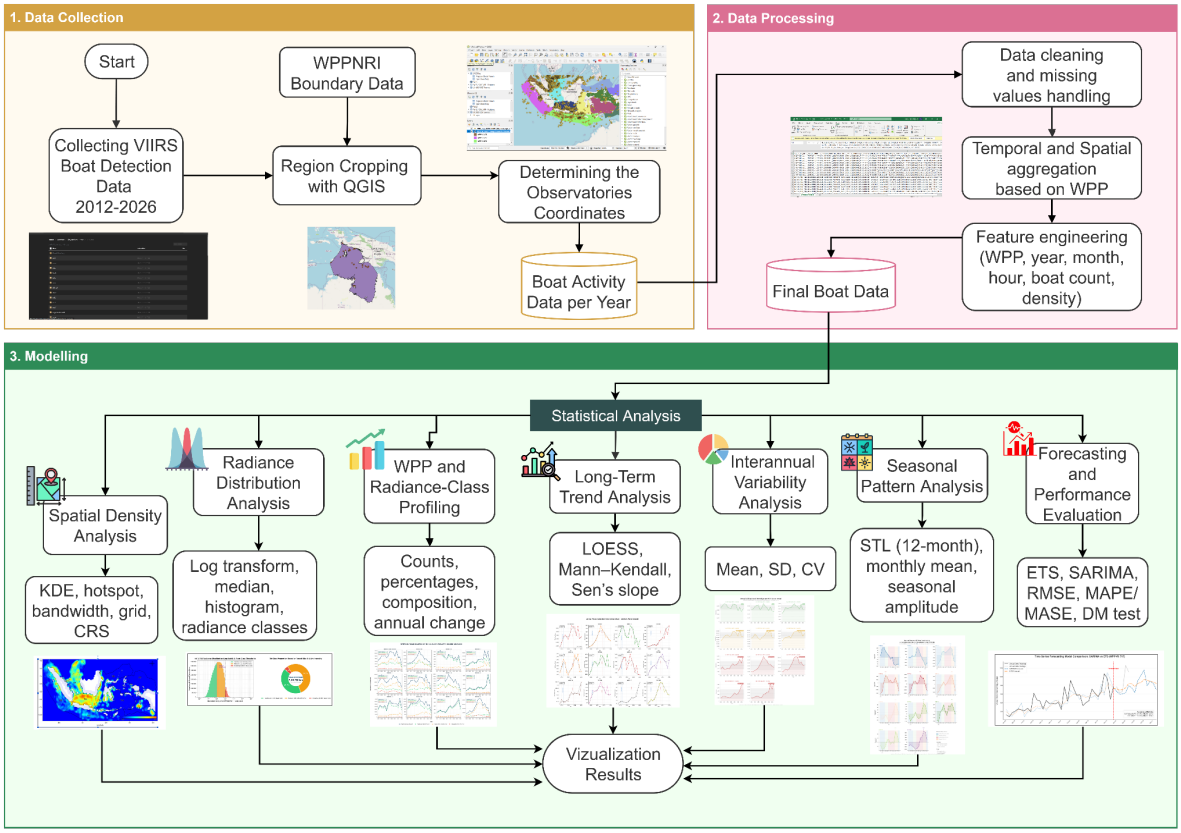


Figure 2. Computational model framework for data processing and statistical analysis of nighttime fishing vessel detections based on VIIRS DNB satellite data.

The computational model was designed so that each technique produces a directly traceable output. The KDE map explains the location of relative concentrations of detection events; the histogram and radiance classes describe the structure of light intensity; the WPP RI profile measures differences in regional composition; the Mann-Kendall test and Sen's slope test long-term trends; the coefficient of variation measures interannual variability; STL extracts the seasonal component; while ETS and SARIMA assess predictive capability for WPP RI with contrasting detection volumes. With this structure, Statistical Analysis in the diagram becomes the node that links the final dataset to all resulting visualizations and tables.

2.3. Data Pre-processing

Pre-processing was carried out using Python for tabular data management and QGIS for geospatial operations. All CSV files were inventoried by year and date, then read incrementally so that the large data volume could be handled efficiently. Column names, data

types, time formats, coordinates, and units were standardized prior to merging. *Date_Mscan* was converted to datetime type and used to construct year, month, date, and hour variables; *Lat_DNB* and *Lon_DNB* were converted to numeric type; while *Rad_DNB* was retained in units of $nW\ cm^{-2}sr^{-1}$.

The initial checks covered file readability, missing values in core variables, coordinates outside the valid geographic range, and duplicate records based on scan time, coordinates, radiance, and quality code. Records lacking a processable time, coordinate, or radiance value were not carried forward to the analysis stage. The primary quality filter set *QF_Detect = 1*, corresponding to the boat category in the EOG VBD product, while *Land_Mask* was used to exclude detections occurring on land. QF1 status was treated as a detection-quality classification assigned by the EOG algorithm, not as verification that every record was definitely associated with a fishing event. This cleaning and reduction process was shown to substantially improve computational efficiency. A quantitative overview of storage efficiency, record-count reduction, and memory optimization at each pre-processing stage is presented in Table 2.

Table 2. Dataset Characteristics at Each Cleaning Stage and Storage Efficiency

Parameter	Raw Data	Intermediate Data (Post-Cleaning)	Analytical Dataset	Description
Number of Attributes	50 columns	11 core attribute groups	Core attributes + derived features	39 attributes excluded from the main matrix (78.0%)
Total File Size	20,03 GB (20.515 MB)	3,85 GB (3.938 MB)	-	80.8% storage reduction
Number of Records	18,115,494	14,211,269	7.380.357	After quality control, spatial filtering, and final data construction
Time Coverage	2012-July 2026	2012-July 2026	2012-July 2026	Year 2026 excluded from annual analysis; 2012 completeness verified from archives

After quality filtering, the WPPNRI shapefile containing all regions was separated by WPP code in QGIS; each polygon was then used to clip the VBD points and assign a WPPNRI attribute to records located within the corresponding region. The resulting clipped dataset was enriched with time variables, WPPNRI code, radiance class, detection count, and density measures. The structure and core attribute scheme retained after cleaning are summarized in Table 3.

Table 3. Structure of Core Attribute Groups in the VIIRS VBD Dataset Post-Cleaning

Attribute	Data Type	Role in Analysis
Lat_DNB	float	Latitude of the DNB pixel center and position of the analysis point.
Lon_DNB	float	Longitude of the DNB pixel center and position of the analysis point.

Rad_DNB	float	Light radiance in nW $cm^{-2}sr^{-1}$; used for distribution and intensity classification.
Date_Mscan/Date	datetime	Scan time; basis for constructing year, month, date, and hour variables.
WPP_RI	string	WPPNRI code resulting from spatial join; spatial stratification unit.
QF_Detect	integer	EOG detection class code; only value 1 is retained in the final dataset.
QF_Bitflag	integer	Quality bitmask information for diagnostic checks.
Land_Mask	integer	Attribute differentiating land, coastline, and water bodies for spatial filtering.
EEZ	string	Jurisdictional boundary context attribute; not used independently to infer illegal activity.
MPA	string	Marine Protected Area context attribute; not used independently to infer violations.
Rad_I04 and Glint	float/flag	Diagnostic attributes to assess heat sources or glint/reflections; not a direct measure of fishing activity.

Monthly aggregation was used for seasonal pattern and forecasting analysis, annual aggregation was used for trend and interannual variability analysis, while individual points were retained for cumulative mapping and Kernel Density Estimation (KDE). For comparative analysis, detection events were operationally grouped into low radiance ($Rad_DNB < 10\ nW\ cm^{-2}sr^{-1}$), medium radiance ($10 \leq Rad_DNB \leq 400\ nW\ cm^{-2}sr^{-1}$), and high radiance ($Rad_DNB > 400\ nW\ cm^{-2}sr^{-1}$), adapting the classification of Tsuda et al. (2023). This classification reflects the radiance level received by the sensor, not vessel size, gross tonnage, fleet class, generator capacity, or gear type. All stages were organized as a pipeline that records the sequence of filters, the definitions of derived variables, the time range, and the number of records before and after filtering, so that the analytical results can be traced back to the source data.

2.4. Statistical Analysis

The statistical analysis was organized into seven components identical to the modeling branches shown in Figure 2, each directly linked to the results. Descriptive statistics were reported as counts, percentages, medians, means, standard deviations, and coefficients of variation. Inferential testing used a significance level of 5% (alpha = 0.05), and all trend tests were two-sided unless stated otherwise. Table 4 summarizes the primary technique and the results served by each component.

Table 4. Alignment Between Analytical Components and Primary Techniques

Analytical Component	**Primary Technique**
Spatial Density Analysis	KDE, shared bandwidth, grid, metric CRS, global scale
Radiance Distribution Analysis	Logarithmic axis, median, histogram, radiance classes
WPP and Radiance-Class Profiling	Counts, percentages, composition, annual change
Long-Term Trend Analysis	LOESS, Mann-Kendall, Sen's slope
Interannual Variability Analysis	Mean, SD, CV
Seasonal Pattern Analysis	STL (12 months), monthly mean, seasonal amplitude

Analytical Component	Primary Technique
Forecasting and Performance Evaluation	ETS, SARIMA, RMSE, MAPE/MASE, DM test

2.4.1. Spatial Density Analysis

Spatial density analysis was used to convert the set of detection points into a continuous surface showing areas of relative concentration. Kernel Density Estimation (KDE) computes the contribution of each point to the surrounding grid cells based on a kernel function and a search radius, or bandwidth. Prior to KDE, points were projected to a metric CRS; this step was necessary so that bandwidth and pixel size would carry a consistent distance meaning. Bandwidth, grid resolution, extent, and the kernel function were kept consistent across WPPs and radiance classes so that pattern changes could not be attributed to changes in map parameters.

KDE was applied to all detection events and subsequently computed separately for the low, medium, and high radiance classes. Areas with relatively high KDE values are referred to as descriptive hotspots, that is, zones of concentrated detection events, rather than as the result of an inferential Getis-Ord hotspot test. To allow comparison across classes, the KDE surfaces were computed using the same bandwidth, grid resolution, extent, kernel function, and CRS. For display purposes, KDE values could be transformed using log1p and then normalized using the global minimum and maximum across the three panels so that a single shared color scale could be used. This transformation was applied only for visualization; comparisons of absolute counts continued to use the raw detection frequency. The KDE approach for identifying the concentration of lit fishing has also been used in West Sumatra waters (Nurholis et al., 2020).

2.4.2. Radiance Distribution Analysis

Radiance distribution analysis aims to characterize the light intensity underlying the detection pattern of nighttime fishing activity. Because *Rad_DNB* has a right-skewed distribution, the histogram of raw values is displayed using a logarithmic x-axis so that both the low and high radiance ranges remain legible, while the y-axis continues to show the frequency of detection events. Observations with non-positive radiance, if any, were excluded from the logarithmic display. The primary measure of central tendency is the median on the original radiance scale, as it is more robust to extreme values than the mean.

The frequency and percentage of the low, medium, and high radiance classes were calculated against all valid detection events using the thresholds of 10 and 400 $nW\,cm^{-2}\,sr^{-1}$. Both thresholds were overlaid on the histogram so that their position relative to the empirical distribution could be assessed. This analysis does not convert radiance into vessel

size or electrical power capacity. Radiance values can be influenced by the number, type, power, orientation, and shielding of lamps; atmospheric conditions and lunar phase; and the position of one or more light sources within the sensor pixel (Elvidge et al., 2015; Hsu et al., 2019; Tsuda et al., 2023).

2.4.3. WPP and Radiance-Class Profiling

The WPP profile was constructed by tabulating the number of detections, the percentage of the overall total, the median radiance, and the composition of radiance classes for each WPP RI. The percentage contribution of a region was calculated as the number of detections in WPP j divided by the total number of detections across all WPPs, multiplied by 100%. Class composition was calculated relative to the total detections within the same WPP so that it could be compared even though observation volumes differ across regions.

$$\text{Percentage WPP}_j = \left(\frac{n_j}{N}\right) \times 100\%$$

$$\text{Class composition } c \text{ for WPP}_j = \left(\frac{n_{j,c}}{n_j}\right) \times 100\%$$

Annual change was calculated from the detection frequency for each WPP RI and radiance class. The complete 2012 to 2025 year series was used to compare compositional change and identify regions that increased, decreased, or fluctuated. Data from January to July 2026 can only be shown as year-to-date (YTD), with a separate symbol and note, and are not compared directly with full years. All profiles represent a measure of detected light activity rather than a count of unique vessels. Differences in frequency may reflect a combination of changes in the number of operating units, the number of nights fished, the intensity of lamp use, and satellite observation opportunities.

2.4.4. Long-Term Trend Analysis

Long-term trends were analyzed using the annual detection counts for the 2012 to 2025 period after checking observation completeness by year. The Locally Estimated Scatterplot Smoothing (LOESS) curve was used as a visual tool to show the shape of nonlinear change without imposing a specific trend function. LOESS was not used as the basis for significance decisions; inferential decisions were obtained from the Mann-Kendall test and Sen's slope estimation.

The Mann-Kendall test was selected because it is nonparametric, does not require a normal distribution, and is suitable for testing monotonic tendencies in environmental time series. The null hypothesis states that no monotonic trend is present, whereas the alternative hypothesis states that an increasing or decreasing trend is present. Trend direction was determined from the sign of Kendall's tau (τ), with significance set at $p < 0.05$ (Mann, 1945).

The magnitude of change was estimated using Sen's slope, the median of the slopes of all pairs of observations, reported in units of detections per year (Sen, 1968).

The results were classified as significantly increasing ($\tau > 0;\ p < 0.05$), significantly decreasing ($\tau < 0;\ p < 0.05$), or showing no evidence of a monotonic trend ($p \geq 0.05$). The last category does not imply that detection counts were truly constant, as the series may still fluctuate or exhibit a nonlinear pattern visible in the LOESS curve. Data from 2026 were excluded from the test so that the partial year would not produce a spurious negative slope.

2.4.5. Interannual Variability Analysis

Interannual variability was calculated from the annual detection count series for 2012 to 2025 after checking observation completeness for each WPP. Three measures were used: the mean, the sample standard deviation (SD), and the coefficient of variation (CV). CV standardizes the standard deviation relative to the mean, allowing the stability of regions with different detection-count scales to be compared. The CV value was calculated as follows.

$$CV = \left(\frac{SD}{\text{mean}}\right) \times 100\%$$

For operational interpretation, WPPs were grouped as stable (CV < 30%), moderately variable (30% ≤ CV ≤ 50%), and highly variable (CV > 50%). These thresholds represent an analytical category defined for this study, not a universal biological standard. A high CV indicates a relatively large change between years but does not by itself explain the cause of that change; its interpretation should be read together with trend direction, radiance composition, policy context, and satellite observation conditions.

2.4.6. Seasonal Pattern Analysis

Seasonal patterns were analyzed using monthly detection counts for complete years. The time series for each WPP was decomposed using Seasonal-Trend decomposition using LOESS (STL) into trend (T_t), seasonal (S_t), and remainder (e_t) components. Because the data have monthly frequency, the seasonal period was set to 12 months (Cleveland et al., 1990). The additive form of the decomposition is expressed simply as follows.

$$Y_t = T_t + S_t + e_t$$

The seasonal component was used to determine the peak and minimum months for each WPP. As a complement to STL, the monthly mean was calculated across all complete years so that calendar patterns could be readily compared. Seasonal amplitude is defined as the difference between the maximum and minimum seasonal values within a single 12-month cycle. The year 2026 was not included in the seasonal pattern estimation because data were available only through July.

$$\text{Seasonal amplitude} = \max(S_1, \ldots, S_{12}) - \min(S_1, \ldots, S_{12})$$

Monthly peaks or troughs are interpreted as patterns in the detection of light-assisted fishing activity. These patterns may relate to the fishing season, monsoon, sea conditions, and operational strategy, but may also be influenced by cloud cover and moon illumination. Attribution to specific oceanographic processes is therefore made in the discussion section with support from relevant references or environmental variables, rather than from STL alone.

2.4.7. Forecasting and Performance Evaluation

Forecasting was performed on the monthly detection count series for two WPPs with contrasting characteristics: WPP RI 712, a region with a very high detection volume, and WPP RI 717, a region with a relatively low detection count. This design was used to test model robustness under volatile conditions and under sparse-data conditions. The data were split chronologically: the period before 2025 was used as training data, while the 12 months of 2025 were used as test data. The test data were not involved in the initial parameter estimation, so that accuracy reflects the model's ability to predict observations it had not seen.

Two model families were compared. The Error-Trend-Seasonal Exponential Smoothing (ETS) model represents a state-space approach that updates the level, trend, and seasonal pattern through exponential weighting. Seasonal Autoregressive Integrated Moving Average (SARIMA) models autoregressive dependence, moving average, differencing, and seasonal structure with a 12-month period. The combination of ETS components and the SARIMA order were selected using only the training data, taking into account residual fit and the corrected information criterion (AICc), so that model complexity was not determined from the test data (Hyndman et al., 2002; Hyndman and Khandakar, 2008).

Accuracy on the test data was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Scaled Error (MASE). RMSE imposes a larger penalty on extreme errors. MAPE is easy to interpret as a percentage but becomes unstable when the actual value is equal to or close to zero; under such conditions, MASE becomes the primary measure because it compares the model's error against the error of a seasonal naive forecast. Smaller values indicate lower prediction error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$$

$$MASE = \frac{\text{mean}|y_t - \hat{y}_t|}{\text{mean}|y_t - y_{t-12}|}$$

The difference in accuracy between ETS and SARIMA was further tested using the Diebold-Mariano (DM) test on the paired loss series. The null hypothesis states that the two models have equivalent predictive accuracy. If $p < 0.05$, the difference in accuracy is declared significant; if $p \geq 0.05$, the model with lower error is described only as descriptively better and is not claimed to be statistically superior (Diebold and Mariano, 1995). Following evaluation, the selected model was refitted on the series available through July 2026 to generate projections for the remaining months of 2026 through early 2027, together with a 95% prediction interval. Actual observations from January to July 2026 were visually distinguished from the forecast values so that actual and projected data would not be conflated.

3. Results and Discussion

This chapter presents the results of a comprehensive analysis of nighttime fishing vessel detection data based on VIIRS DNB satellite observations across all WPPNRI during the 2012 to 2026 period. Consistent with the study's methodological flow, the presentation of results and discussion in this chapter is organized hierarchically into four main stages of analysis. First, an analysis of the spatial profile and fleet characteristics, focusing on activity density (kernel density) mapping and the classification of vessel operational scale based on light intensity (radiance). Second, an evaluation of long-term dynamics and trend stability using the Mann-Kendall statistical test and variability indices to track shifts in fishing grounds. Third, time-series decomposition to identify seasonal fluctuation patterns and fleet activity responses to monsoon climate dynamics. Finally, validation of predictive forecasting models using the SARIMA and ETS algorithms in waters with extreme characteristics, to test the reliability of a data-based early warning system for future fisheries governance.

3.1. Spatial Profile and Fleet Characteristics of Indonesia's Nighttime Capture Fisheries

Spatio-temporal analysis of nighttime fishing activity in Indonesian waters over the 2012 to 2026 period reveals a highly heterogeneous fleet operational structure across WPPNRI. Based on the KDE Spatial Density Map in Figure 3, light-fishing vessel activity is not evenly distributed but instead concentrated in several main epicenters (hotspots). The highest absolute density is centered in the shallow epicontinental waters of the western region, principally the Java Sea (WPP 712), the Malacca Strait (WPP 571), and the Natuna Sea (WPP 711), as well as the eastern water corridors of the Arafura Sea (WPP 718) and the Makassar Strait (WPP 713). In contrast, the oceanic and deep-sea waters of the eastern region, such as Cenderawasih Bay (WPP 717), the Halmahera Sea (WPP 715), and the Indian Ocean west of Sumatra (WPP 572), record a much lower point density (sparse data regions)

with a dispersed distribution pattern. This pattern confirms the preference of lit fishing fleets to operate in nutrient-rich, productive neritic zones (Sarangi & Jaiganesh, 2021).

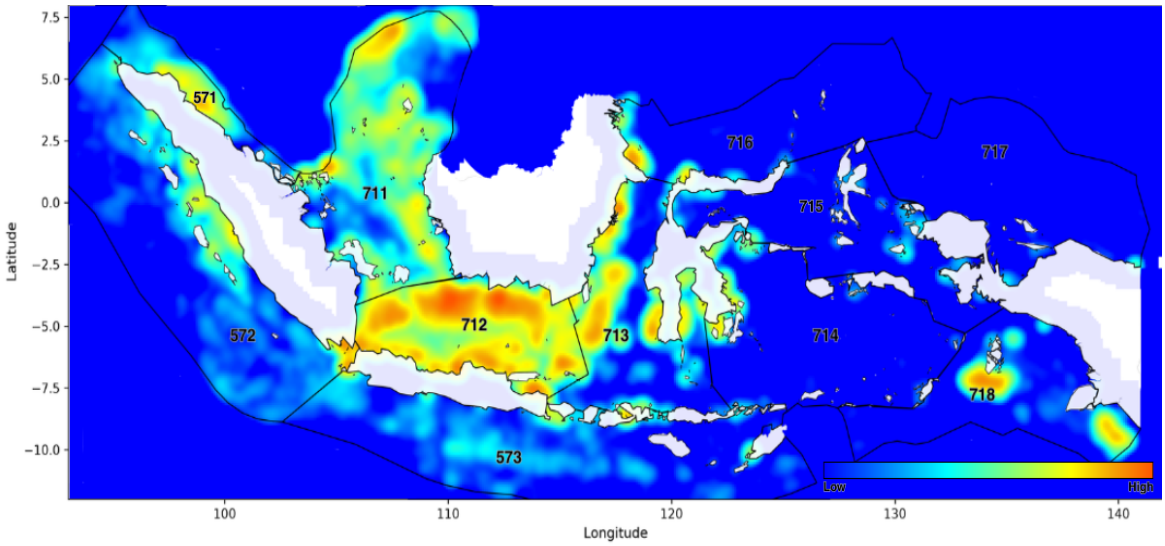


Figure 3. Spatial Density Map (Kernel Density Estimation - KDE) of Nighttime Fishing Vessel Detections Across Indonesian Fisheries Management Areas (WPPNRI).

To understand the fleet characteristics underlying this spatial density pattern, the extraction of radiance emissions from the VIIRS DNB sensor yielded a cumulative total of 7,380,357 valid lit-vessel detections. As shown in Figure 4a, the radiance distribution (logarithmic scale), with a median value of 13.63 nW/cm2 sr, was used to classify the fleet into three operational-scale classes. This grouping follows the sensitivity standard for vessel lamp output power, namely the Traditional/Small-Scale Fleet (less than 10 nW cm^{-2} sr^{-1}), the Medium-Scale Fleet (10 to 400 nW cm^{-2} sr^{-1}), and the Industrial-Scale Fleet (over than 400 nW cm^{-2} sr^{-1}) (Tsuda et al., 2023). In aggregate, as shown in Figure 4b, the structure of Indonesia's nighttime fisheries is overwhelmingly dominated by the combination of the medium-scale fleet (53.7%) and the traditional/small-scale fleet (41.8%). By contrast, the very high-power industrial fleet segment accounts for only a minority share of 4.5% of the total detection volume. This dominance is consistent with the general characteristics of tropical fisheries (multi-species and multi-gear), which rely on small to medium artisanal fleets based on light attraction, such as fixed or floating lift nets and small pelagic purse seines (FAO, 2022; Herdiana et al., 2024).

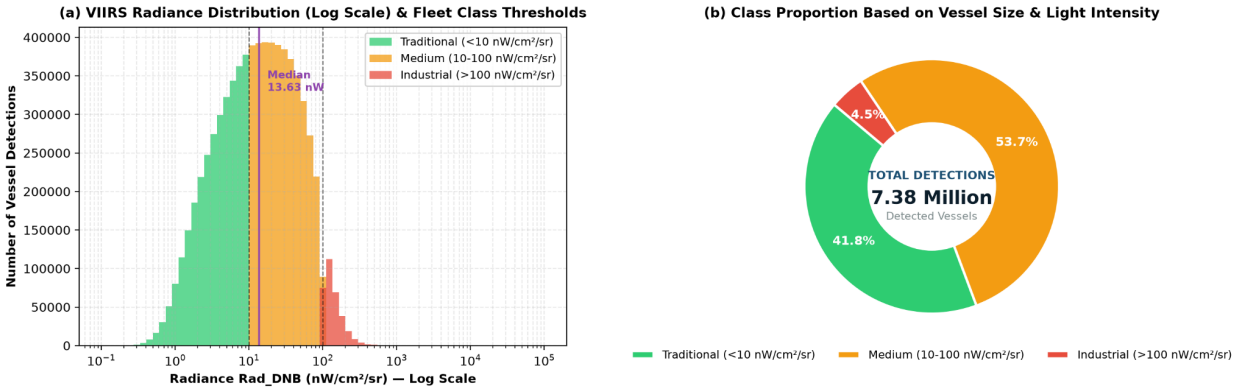


Figure 4. (a) Log-scaled Radiance Distribution and Fleet Classification Thresholds; (b) Overall Proportion of Vessel Classes Across All WPPNRI Based on Radiance Intensity (2012-2026)

Although the medium-scale fleet dominates in aggregate across Indonesian waters, the regional breakdown of fleet-class proportions in Figure 5 and Table 5 reveals a sharp spatial dichotomy. WPP 712 (Java Sea) confirms its position as the largest activity center, accounting for 3.88 million detections, or 52.7% of the total accumulation across all WPPNRI. The density in the Java Sea is driven by the dominance of medium (61.6%) and traditional (33.9%) fleets. The shallow topography of the Java Sea, together with rich terrestrial nutrient input, creates an ideal habitat for small pelagic commodities such as scad (Decapterus spp.) and sardinella (Sardinella spp.) (Nabil et al., 2024). Historically, these conditions have driven the concentration of light-assisted purse seine operations, which carry a high risk of contributing to excess capacity (Squires et al., 2003; Purwanto et al., 2022).

A contrasting phenomenon is observed at the maritime border and offshore waters, particularly WPP 711 (Natuna Sea) and WPP 718 (Arafura Sea). WPP 711 recorded the highest proportion of the industrial-scale fleet nationally, reaching 12.0% (64,080 detections). This high share of the industrial class in these transboundary waters indicates the dominance of squid jigging operations and large, high-power pelagic purse seining. This finding is consistent with the current state of the art in global satellite observation regarding the use of high-intensity lamp power to attract squid and tuna across international borders (Park et al., 2020; Wang et al., 2025).

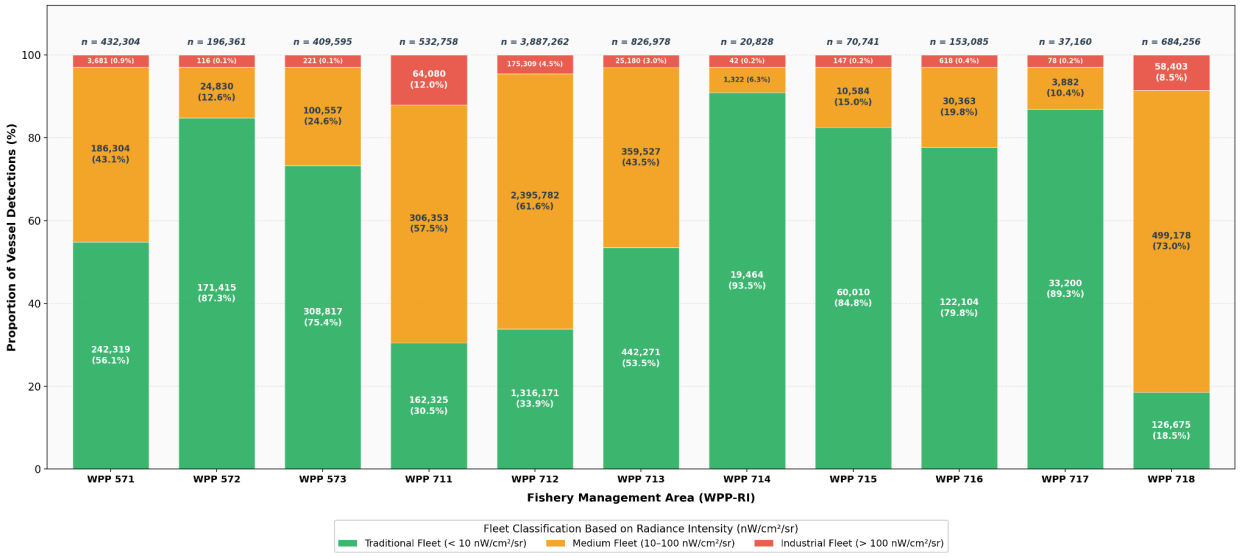


Figure 5. Composition of Vessel Classes (Traditional, Medium, Industrial) across 11 Indonesian Fisheries Management Areas (WPP-RI).

By contrast, eastern and open-ocean regions such as WPP 714 (Banda Sea), WPP 717 (Cenderawasih Bay), and WPP 572 (Indian Ocean west of Sumatra) are dominated by the traditional/small-scale fleet, with proportions reaching 93.5%, 89.3%, and 87.3%, respectively. The dominance of low-power fleets (less than 10 nW/cm2 sr) in these regions indicates that fishing activity remains largely subsistence or artisanal in nature, with limited large commercial fleets (Sondita et al., 2022). Beyond fleet-related factors, the low detection volume in the eastern region is also influenced by technical challenges, namely high seasonal cloud cover, which reduces the penetration sensitivity of the VIIRS DNB sensor (Elvidge et al., 2021).

Overall, the combined evidence in Figures 3, 4, and 5 demonstrates that Indonesia's nighttime fisheries are divided into two main categories: (1) a coastal-epipelagic category in the highly dense western region, dominated by small to medium fleets, and (2) an industrial-frontier category in the border and offshore regions, characterized by high light-energy intensity.

Table 5. Summary Profile of WPPNRI: Detection Volume, Radiance Intensity, and Practical Fisheries Implications (2012-2026)

WPPNRI Region	Detections (n)	Share (%)	Median (nW)	Low (%)	Medium (%)	High (%)	Key Characteristics & Practical Fisheries Implications
WPP 571	432,304	5.9	8.09	56.1%	43.1	0.9%	Relatively high detection activity in the Malacca Strait, dominated by low and medium radiance classes.
WPP 572	196,361	2.7	3.03	87.3%	12.6	0.1%	Predominantly low-radiance detections in the Indian Ocean West of Sumatra.
WPP 573	409,597	5.5	4.81	75.4	24.6	0.1	Dominated by low-radiance detections, with a relatively higher contribution of medium radiance compared to WPPNRI 572.
WPP 711	532,758	7.2	22.87	30.5	57.5	12.0	Features the largest proportion of high-radiance detections, although the medium radiance class remains dominant.
WPP 712	3,887,263	52.7	17.21	33.9	61.6	4.5	The largest detection hub across all WPPNRI, with a strong dominance of medium radiance.
WPP 713	826,975	11.2	8.67	53.5	43.5	3.0	High detection volume with a relatively balanced composition of low and medium radiance.
WPP 714	20,828	0.3	2.54	93.5	6.3	0.2	Strongly dominated by low-radiance detections with a relatively small overall detection volume.
WPP 715	70,741	1.0	3.57	84.8	15.0	0.2	Dominated by low-radiance detections in Maluku and its surrounding waters.
WPP 716	153,085	2.1	4.23	79.8	19.8	0.4	Dominated by low-radiance detections, with a minor contribution from medium radiance.
WPP 717	37,160	0.5	2.73	89.3	10.4	0.2	Lowest total detection volume across all WPPNRI, predominantly low radiance.
WPP 718	684,256	9.3	30.78	18.5	73.0	8.5	Exhibits the highest median radiance and strong medium-radiance dominance, with a

WPPNRI Region	Detections (n)	Share (%)	Median (nW)	Low (%)	Medium (%)	High (%)	Key Characteristics & Practical Fisheries Implications
							relatively substantial high-radiance proportion.
Total	**7,380,357**	**100.0**	**13.63**	**41.8**	**53.7**	**4.5**	**Across all Indonesian waters, detections are predominantly dominated by medium- and low-radiance classes.**

This difference in radiance composition is more clearly reflected in its geographic distribution pattern. Referring to the spatial distribution in Figure 6 and the data summary in Table 5, each class shows a different level of concentration in each WPP RI. WPP 712 is the largest activity center (52.7%), followed by WPP 713 (11.2%), WPP 718 (9.3%), and WPP 711 (7.2%). Together, these four strategic regions account for approximately 80.4% of total detections across the entire study area. These closely clustered points form hotspots of differing intensity within each radiance class.

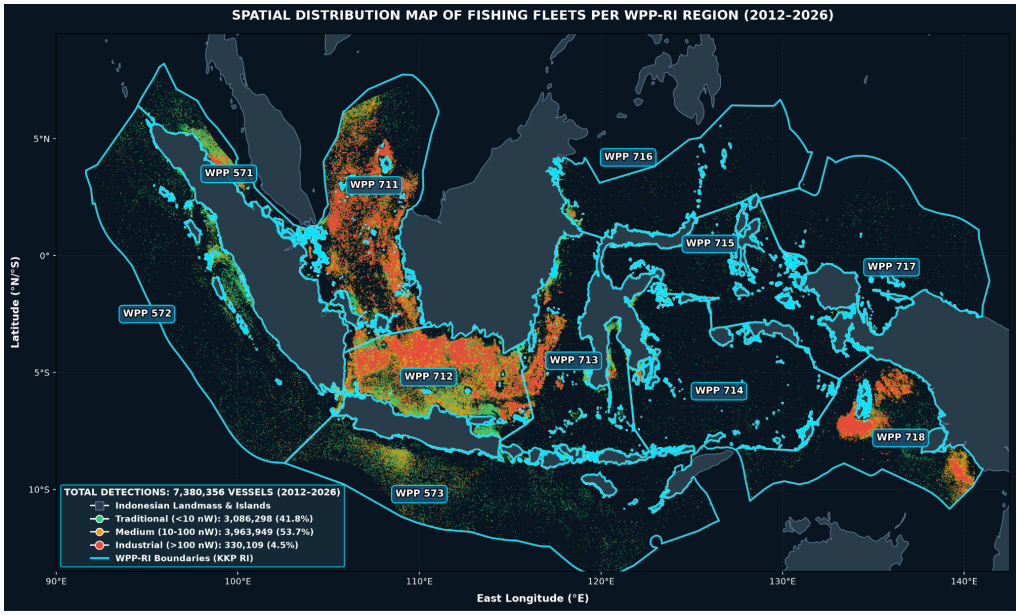


Figure 6. Spatial distribution of VIIRS Boat Detections by radiance class across the 11 WPPNRI regions (2012-2026).

This density pattern is shown in greater detail in Figure 7. The density map shows that the low and medium radiance classes have a relatively broad distribution, with the main concentrations in WPP 571, WPP 711, WPP 712, and WPP 713. In contrast, the high radiance class shows a much more localized pattern, particularly in WPP 711, WPP 712, and WPP 718. It should be noted that the density scale in each panel was normalized separately, so color intensity can only be compared within the same radiance class.

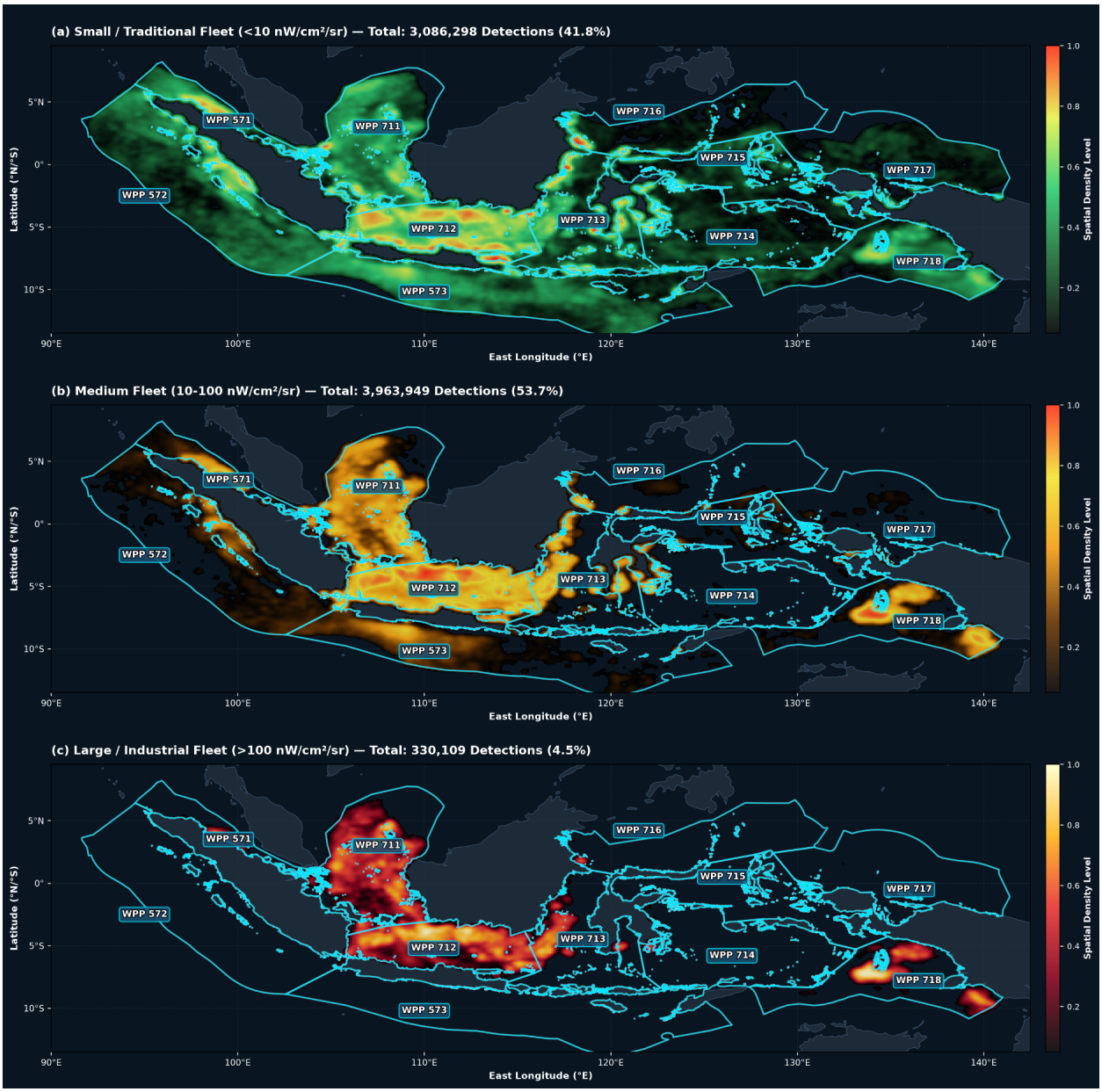


Figure 7. Spatial Density Heatmap of Fishing Vessels per Fleet Class

Figure 7 shows the spatial density of VIIRS Boat Detection events by low, medium, and high radiance class across the 11 WPP RI during the 2012 to 2026 period. The density map in Figure 7 shows that the low and medium radiance classes have a relatively broad distribution, with the main concentrations in WPP 571, WPP 711, WPP 712, and WPP 713. In contrast, the high radiance class shows a more localized pattern, particularly in WPP 711, WPP 712, and WPP 718. The density scale in each panel was normalized separately, so color intensity can only be compared within the same radiance class.

This spatial concentration is not static, as detection counts in each WPP RI change from year to year. The dynamics of the low, medium, and high radiance classes over the observation period are presented in Figure 8.

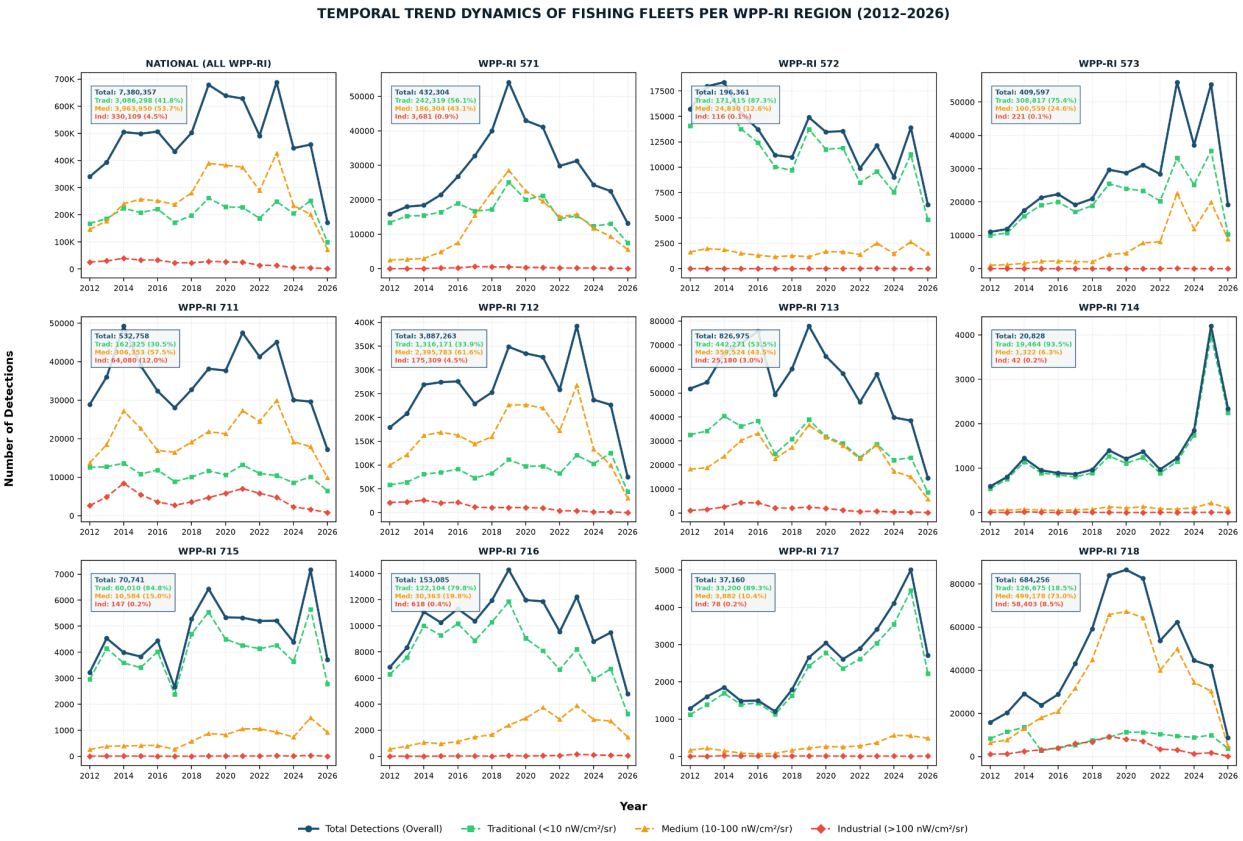


Figure 8. Annual dynamics of VIIRS Boat Detection counts by radiance class overall and across individual WPPNRI regions (2012-2026).

This spatial concentration is dynamic, as detection counts in each WPP RI change from year to year. The temporal dynamics of the low, medium, and high radiance classes over the observation period, shown in Figure 8, indicate that activity change did not occur uniformly. WPP 712 has the largest detection volume but shows very strong annual fluctuation. WPP 573 and WPP 718, on the other hand, show a long-term growth tendency in activity, whereas WPP 572 instead shows a declining tendency. It should be noted that the sharp decline visible at the end of the 2026 graph occurs because the 2026 data represent an in-progress year that does not yet represent a full year of observation, so the extreme fluctuation in that year was not included in the statistical trend test.

As a baseline conclusion, lit fishing vessel activity in Indonesia is dominated by the low-to-medium radiance classes and is strongly concentrated in WPP 712. Nevertheless, the radiance composition, hotspot locations, and annual change patterns show highly specific

characteristics for each WPP RI. This spatial baseline mapping and the differences in temporal dynamics are further analyzed statistically using the Mann-Kendall test and Sen's slope estimation to identify the significance of long-term trend shifts.

3.2 Long-Term Dynamics and Fleet Activity Stability Across Fisheries Management Areas (WPP RI)

An evaluation of the temporal trajectory over the 2012 to 2025 period reveals a dynamic shift in nighttime fishing exploitation patterns across Indonesian waters. Based on the results of the non-parametric Mann-Kendall test and Sen's Slope estimation, the growth trend in the number of lit fishing vessels shows a highly significant spatial polarization ($p < 0.05$) between the western and eastern regions of Indonesian waters. The most prominent phenomenon emerging from this long-term trend analysis is the accelerating expansion of fleets in eastern Indonesia (Eastern Frontier Expansion). As shown in the long-term temporal trend in Figure 9, consistent and statistically significant fleet growth ($p < 0.05$) was recorded in WPP 714 (Banda Sea), WPP 715 (Halmahera/Seram Sea), WPP 717 (Cenderawasih Bay), and WPP 718 (Arafura Sea). This geographic shift in fishing activity provides quantitative evidence of the spatial reorganization of medium-scale and industrial fleets. This reorganization was strongly influenced by the Ex-Foreign Vessel Moratorium policy enacted in late 2014, clearly recorded in the curve as the Moratorium line, which triggered a temporary decline in operations before local fleets ultimately expanded into the eastern frontier waters (Rochman et al., 2021).

In contrast to the rapid acceleration in the eastern region, the traditional fishing waters of western Indonesia show indications of saturation and an ecological carrying-capacity limit. The Mann-Kendall test for WPP 712 (Java Sea) and WPP 571 (Malacca Strait) confirms a Stagnant or No-Trend status ($p > 0.05$). Although WPP 712 contributes the largest annual detection volume nationally, the LOESS fitted curve in Figure 9 shows a sharp boom-and-bust fluctuation pattern without any meaningful long-term increase in the mean. This plateau pattern suggests that the Java Sea may have reached the maximum absorptive capacity of its ecosystem (excess capacity), where additional lit fishing units no longer produce a sustained increase in detection counts (Squires et al., 2003; Purwanto et al., 2022). A more critical condition is observed in WPP 572 (Indian Ocean west of Sumatra), the only region showing a consistent and significant declining fleet trend ($p < 0.05$). This declining trend likely reflects oceanographic dynamics in open-ocean waters that are less conducive to small to medium-scale vessel operations, along with a shift of activity to other regions (Muawanah et al., 2021).

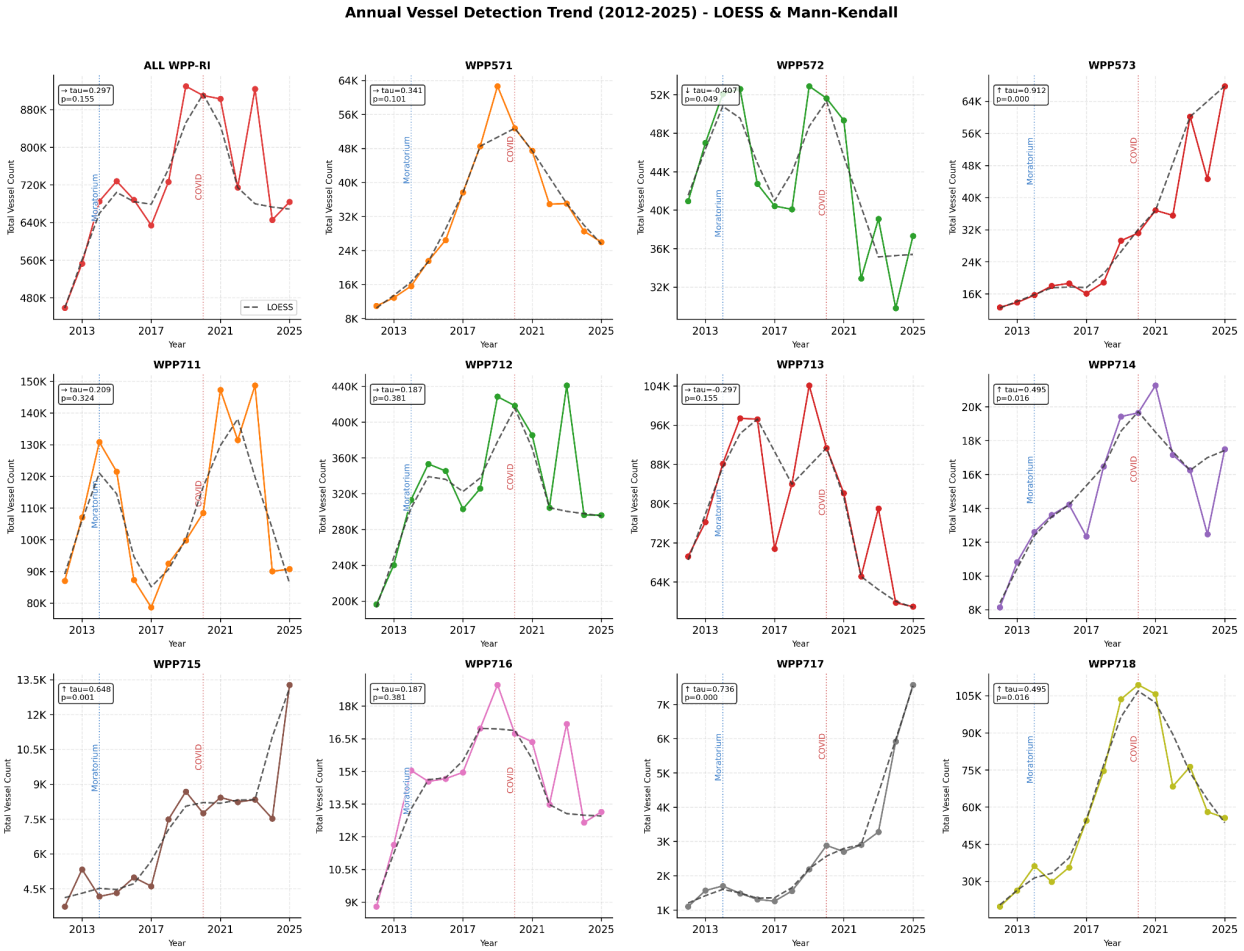


Figure 9. Annual Vessel Detection Trend (2012-2025) across 11 WPP-RI and the Overall Study Area Aggregate, Overlaid with LOESS Smoothing Curve and Mann-Kendall Trend Test.

This dynamic structure is reinforced visually through the trend stability index analysis (Trend Stability Index / Coefficient of Variation) in Figure 10. Based on their level of variability, the WPPNRI are divided into three main stability clusters reflecting different operational characteristics. The first cluster comprises high-stability regions (Stable, CV < 30%), represented by WPP 571 (CV = 26.7%), WPP 712 (CV = 26.4%), and WPP 713 (CV = 25.9%). This low level of variability indicates that the fishing grounds in these waters are relatively saturated with established local fleets, so that additions to the vessel population occur at a fairly constant rate. The second cluster comprises moderately variable regions (Moderately Variable, CV = 30-50%), such as WPP 572, WPP 711, and WPP 714, reflecting transitional areas resulting from the dynamics of operational adjustment and cross-border regulatory oversight. The third cluster consists of highly variable regions (Highly Variable, CV > 50%), represented by rapidly expanding areas such as WPP 573 (CV = 51.2%), WPP 718 (CV = 51.5%), WPP 717 (CV = 63.4%), and WPP 715 (CV = 66.2%). The extreme fluctuation and variability in this cluster are driven by the high sensitivity of expansion

regions to the entry of new fleet investors, the mobilization of vessels between regions, and shifts in regional governance regulations (Solihin et al., 2022).

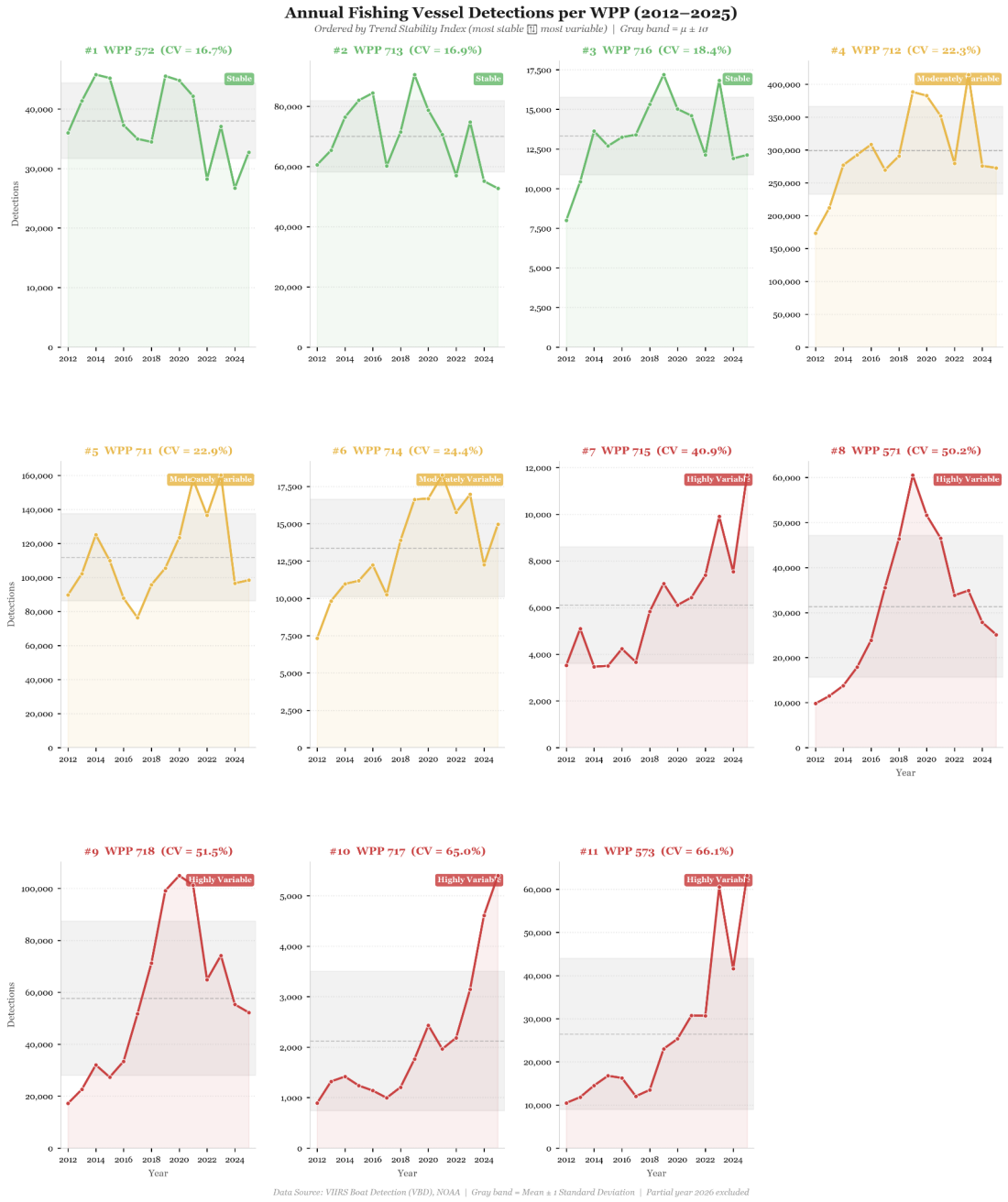


Figure 10. Categorization of WPP-RI Based on Trend Stability Index (Coefficient of Variation - CV%) Ranging from Stable (Green), Moderately Variable (Yellow), to Highly Variable (Red).

Table 6. Mann-Kendall Test Results and Sen's Slope Estimates for Vessel Detection Activity Across 11 WPPNRI Regions (2012-2025)

WPPNRI Region	Kendall's Tau (τ)	p-value	Sen's Slope (detections/year)	Trend Direction	Significance
WPP 571	0.341	0.101	+2,103.0	↔	Not Significant
WPP 572	-0.407	0.049	-875.5	↓	Significant
WPP 573	0.912	<0.001	+2,960.7	↑	Significant
WPP 711	0.209	0.324	+914.0	↔	Not Significant
WPP 712	0.187	0.381	+7,671.3	↔	Not Significant
WPP 713	-0.297	0.155	-1,362.3	↔	Not Significant
WPP 714	0.495	0.016	+650.8	↑	Significant
WPP 715	0.648	0.001	+463.4	↑	Significant
WPP 716	0.187	0.381	+236.8	↔	Not Significant
WPP 717	0.736	<0.001	+278.6	↑	Significant
WPP 718	0.495	0.016	+4,672.6	↑	Significant

Overall, the findings from Table 6, Figure 9, and Figure 10 demonstrate that the growth pattern of Indonesia's nighttime fishing fleet does not follow a uniform linear trajectory, but is instead marked by saturation in western waters and an expansive shift toward the eastern region. This dynamic long-term trend movement was clearly interrupted by external shocks such as the Moratorium policy and the COVID-19 pandemic, underscoring the importance of understanding how seasonal fluctuations and annual climate cycles affect fleet operations more specifically at the local level, which is discussed in the following subsection.

3.3 Seasonal Patterns of Fishing Activity and Response to Monsoon Climate Phenomena

Beyond the long-term growth trajectory, the daily operational dynamics of Indonesia's nighttime fishing fleet are strongly regulated by seasonal rhythms driven by the monsoon climate system. Through Seasonal-Trend Decomposition using LOESS (STL) (Cleveland et al., 1990), the seasonal variable was successfully isolated from the trend and remainder components, revealing how fluctuations in sea weather constrain or promote lit vessel activity. As shown in the seasonal index derived from the STL decomposition in Figure 11 and the monthly mean distribution pattern in Figure 12, most WPPs in central and eastern Indonesia, such as the Java Sea (WPP 712), the Makassar Strait (WPP 713), the Indian Ocean south of Java (WPP 573), and the Arafura Sea (WPP 718), reach their peak activity (peak season) during the August to October period. The activity peak in these regions coincides precisely with the Southeast Monsoon (East Monsoon) phase. The steady trade winds during this phase trigger upwelling, which stimulates a chlorophyll-a bloom and aggregates schools of small pelagic fish and squid (Nababan et al., 2022; Setyohadi et al., 2022; Siregar et al., 2020). Combined with relatively calm seas and low rainfall, these conditions drive the fishing intensity of lit vessel fleets to its annual maximum.

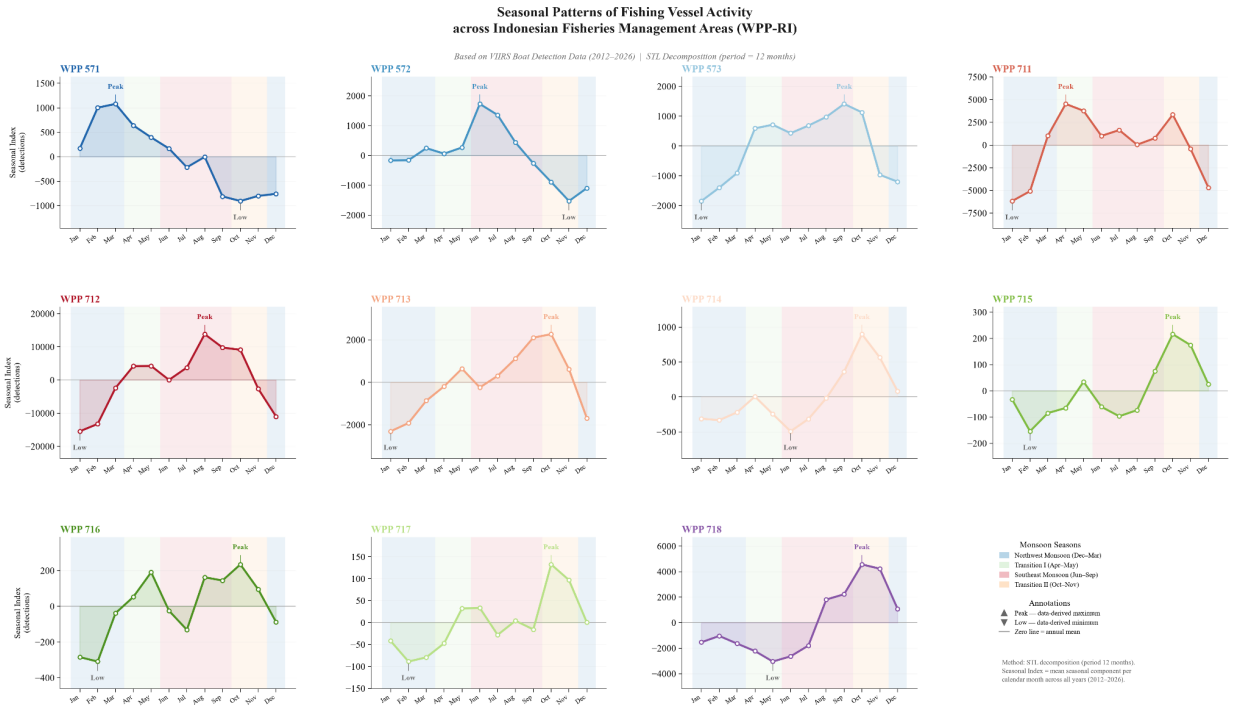


Figure 11. Seasonal Component of Fishing Vessel Detections Extracted via STL Decomposition across 11 WPP-RI, Indicating Monthly Peak Seasons and Seasonal Amplitudes.

Conversely, a sharp decline in the fishing activity index (low season) occurs consistently across most WPPs in the first quarter, particularly from January to February, as a direct response to the peak of the Northwest Monsoon (West Monsoon). During this phase, strong winds from the Asian continent generate high waves, strong surface currents, and extreme rainfall intensity and cloud cover across most of Indonesia's waters. These extreme weather conditions act as a natural ecological barrier that forces fishing operators to halt operations, moor their vessels in port, or carry out gear maintenance (Susanto et al., 2021). Beyond endangering navigational safety, the heavy cloud cover and large waves during the West Monsoon also physically reduce the effectiveness of light penetration from attractor lamps into the water column, thereby significantly lowering technical fishing efficiency (Elvidge et al., 2021).

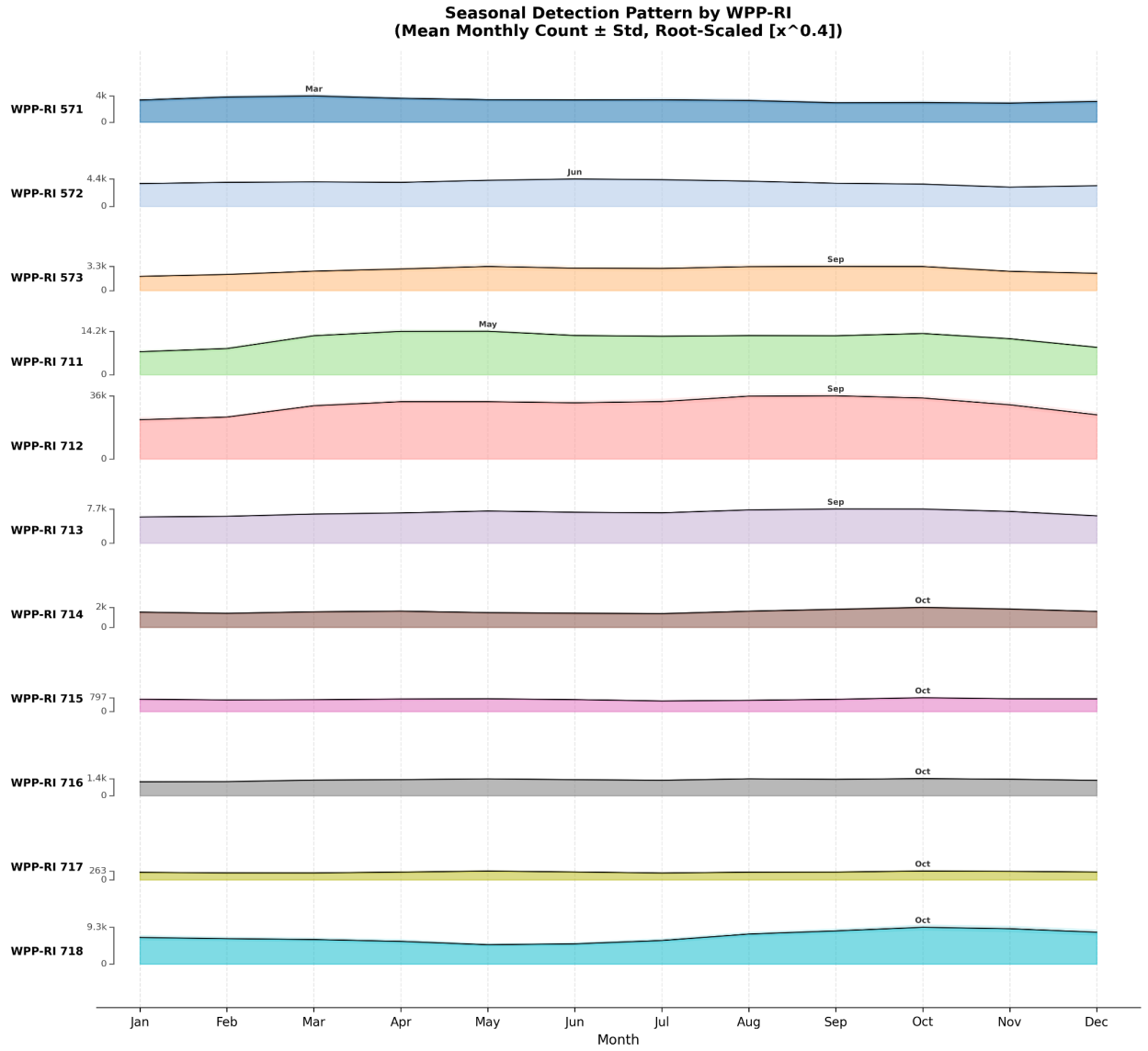


Figure 12. Mean Monthly Detection Distribution across WPP-RI (Root-Scaled [$x^{0.4}$]), Highlighting Inter-regional Seasonal Amplitudes and Shifts in Peak Months.

Although the influence of the monsoon is felt throughout Indonesian waters, Figures 11 and 12 confirm region-specific variation in the shift of peak months and the magnitude of seasonal fluctuation (seasonal amplitude) across WPPs. Unlike the central and eastern regions, western waters such as the Malacca Strait (WPP 571) record an earlier activity peak in March, while the Indian Ocean west of Sumatra (WPP 572) reaches its peak in June. In addition, shallow open waters such as the Java Sea (WPP 712) and the Arafura Sea (WPP 718) show the most sensitive seasonal response, with a very wide fluctuation amplitude resulting from rapid changes in surface thermal structure. In contrast, deep and sheltered waters such as the Halmahera/Seram Sea (WPP 715) and the Sulawesi Sea (WPP 716) show a relatively flat seasonal curve. Deep-sea bathymetry and archipelagic topography provide

natural physical protection against surface wave exposure, allowing fleets in these regions to operate more stably throughout the year. This comprehensive understanding of the relationship between seasonal climate and inter-regional variability demonstrates that future predictive modeling and fisheries quota management must be adapted to the specific seasonal calendar of each WPP.

3.4 Predictive Forecasting Evaluation: Performance Comparison in Regions with Extreme Density (WPP 712 and WPP 717)

The integration of long-term trend understanding and seasonal fluctuation forms the primary foundation for building a reliable predictive forecasting model for data-based fisheries governance. To test the reliability and robustness of the time-series forecasting algorithms, a comparative stress test was conducted under two extreme conditions: WPP 712 (Java Sea) as the densest activity epicenter across the entire study area, and WPP 717 (Cenderawasih Bay) as the region with the lowest activity. The highly contrasting spatio-temporal characteristics of these two regions are illustrated visually through the monthly mapping in Figures 13 and 14. WPP 712 represents hyper-active waters with enormous density, while WPP 717 represents a sparse data region with a very low point density (< 4,500 detections/year). Forecasting across both of these spectrums simultaneously constitutes a critical mathematical test, given that most quantitative algorithms frequently suffer accuracy degradation when faced with either extreme fluctuation volatility (as in WPP 712) or high intermittency patterns (as in WPP 717).

To determine the best predictive model, this study compared the performance of the two most dominant classical time-series analysis algorithms, namely SARIMA (Box et al., 2015) and Error, Trend, Seasonal Exponential Smoothing (ETS) (Hyndman & Athanasopoulos, 2021). Qualitative and quantitative evaluation of the projection test data is presented comparatively in Figure 15. The error-metric evaluation results show that the ETS model consistently outperforms SARIMA across both WPP spectrums. In the low-data region (WPP 717), ETS recorded a Mean Absolute Percentage Error (MAPE) of 17.5% (Good Forecasting category) and a Root Mean Squared Error (RMSE) of 264.2, more robust than SARIMA, which recorded a MAPE of 21.4% and an RMSE of 321.7. A similar advantage occurred in the high-volatility region (WPP 712), where ETS showed superior performance with a MAPE of 24.6% and an RMSE of 8195.9, far outperforming SARIMA, which had an error level of MAPE 37.6% and RMSE 10192.1.

The superior performance of the ETS model under both extreme conditions lies in its adaptive exponential weighting mechanism, which flexibly assigns greater weight to more recent observation dynamics (Kim et al., 2023). This approach allows ETS to track the intermittent real-data fluctuations in WPP 717 responsively, while simultaneously dampening the excessively large variance shocks in WPP 712. In contrast, the inherently linear nature of the SARIMA model, which relies heavily on long-term autoregressive

memory components, exhibits fundamental vulnerability when non-stationary seasonal shifts occur. As a result, SARIMA projections tend to be rigid and inaccurate, either overfitting or underforecasting, when attempting to capture sudden field-level surges (Silva et al., 2021).

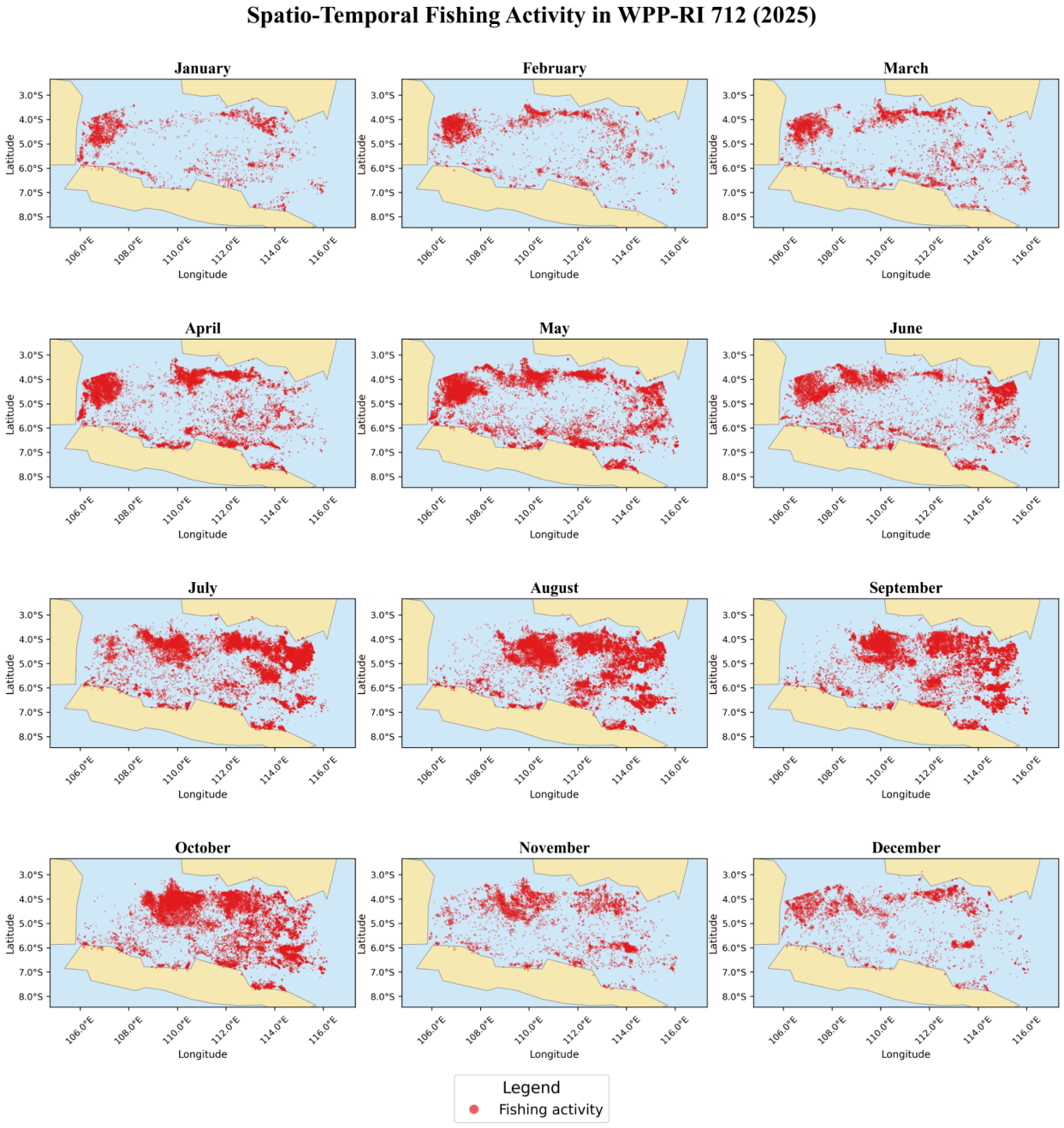


Figure 13. Spatio-Temporal Fishing Activity Distribution in WPP-RI 712 Across 12 Months (2025), Exhibiting Sparse Data Characteristics and Seasonal Shifts in Fishing Grounds.

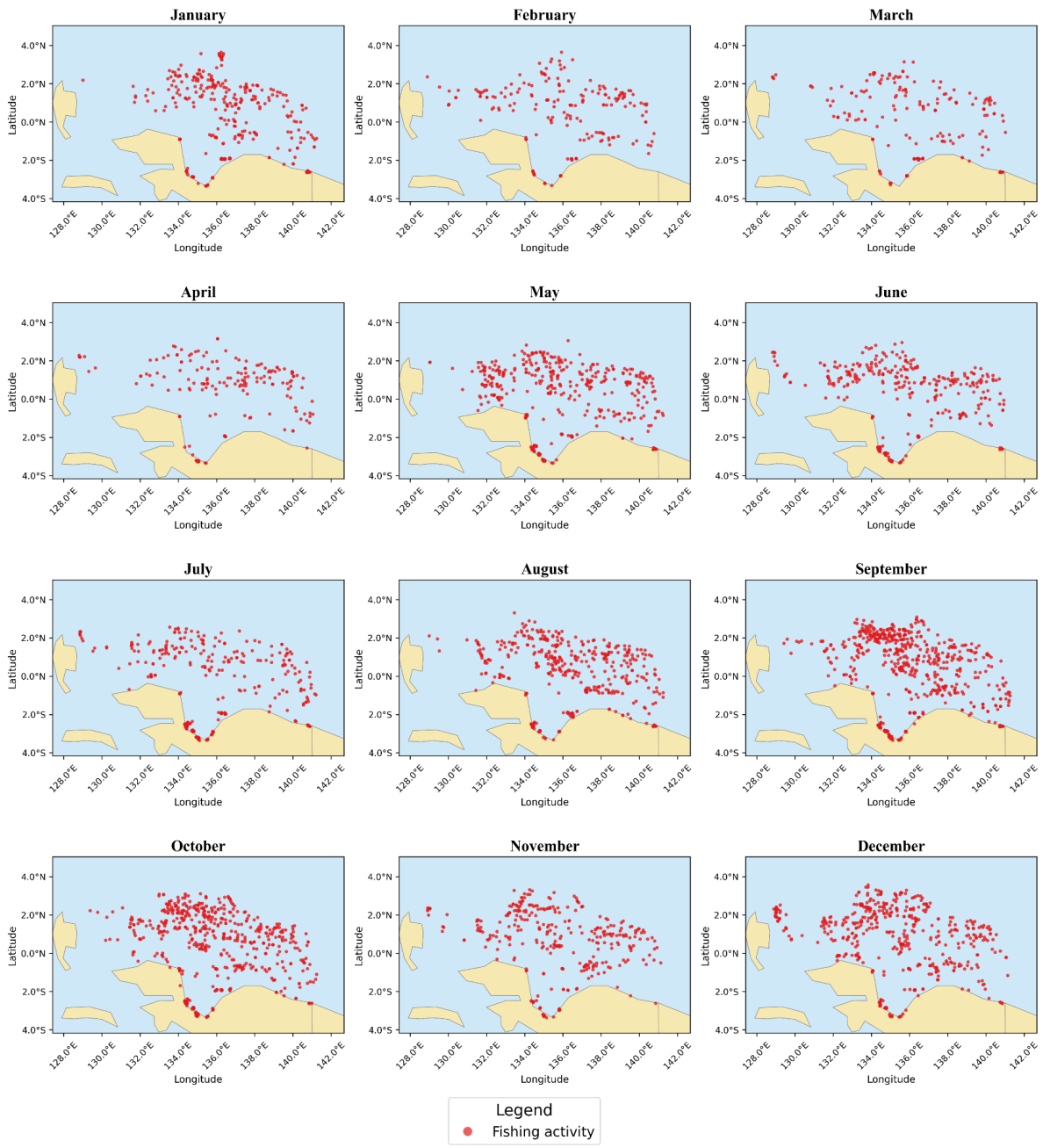


Figure 14. Spatio-Temporal Fishing Activity Distribution in WPP-RI 717 Across 12 Months (2025), Exhibiting Sparse Data Characteristics and Seasonal Shifts in Fishing Grounds.

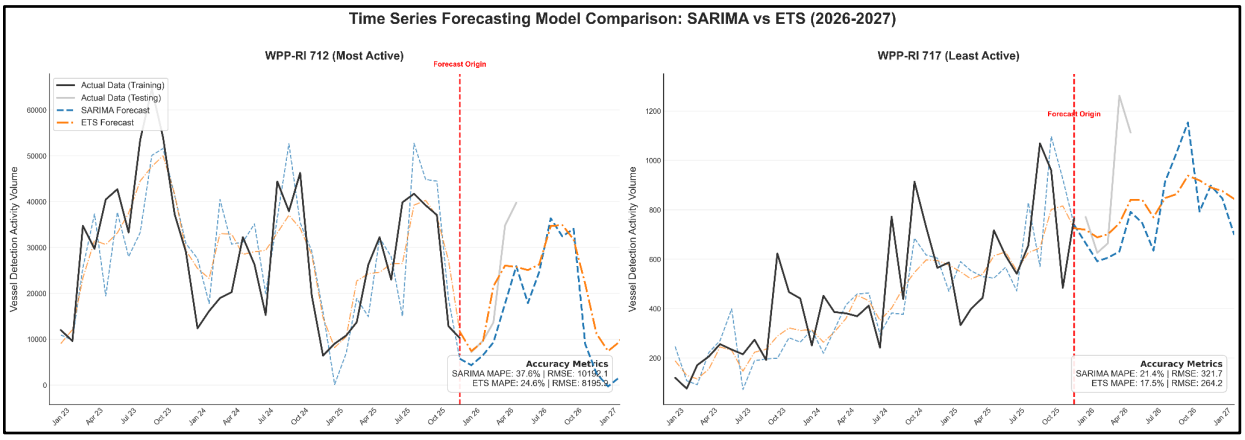


Figure 15. Time Series Forecasting Model Comparison between SARIMA and ETS for WPP-RI 717, Showing Actual Training/Testing Data, Forecast Horizon (2025-2027), and Accuracy Metrics.

The implications of the Diebold-Mariano test results presented in Table 7 indicate that the difference in performance between the ETS and SARIMA models is not yet statistically significant at the alpha = 0.05 level. For WPP 712, the p-value was 0.566 for MAPE and 0.403 for RMSE, while for WPP 717 the p-values were 0.409 for MAPE and 0.116 for RMSE. All p-values exceed the 0.05 significance threshold, so there is insufficient evidence to state that either model has a statistically superior forecasting accuracy compared with the other. Thus, although Figure 15 shows that ETS produced lower MAPE and RMSE values than SARIMA for both WPPs, this advantage is descriptive in nature and has not been shown to be statistically significant based on the Diebold-Mariano test.

Table 7. Diebold-Mariano Test Results for Model Performance Comparison Between SARIMA and ETS

WPP-RI	Loss Function	DM Statistic	p-value	Significance
WPP 712	MAPE	-0.574	0.566	Not Significant
WPP 712	RMSE	+0.836	0.403	Not Significant
WPP 717	MAPE	-0.825	0.409	Not Significant
WPP 717	RMSE	-1.572	0.116	Not Significant

4. Conclusion

This study successfully characterized the spatio-temporal dynamics of light-assisted nighttime fishing activity across the 11 WPPNRI using VIIRS DNB data over the 2012 to 2026 observation period. Detections were dominated by the medium (53.7%) and low (41.8%) radiance classes, with the highest concentration in WPP 712. Trend analysis for 2012 to 2025 revealed a spatial reorganization characterized by stagnation in several western WPPNRI

and increasing activity in several eastern regions, forming an Eastern Frontier Expansion pattern, while the STL analysis showed a strong seasonal pattern, with region-wide activity peaking in the third quarter, coinciding with the Southeast Monsoon. In the forecasting of WPP 712 and WPP 717, ETS produced lower descriptive error than SARIMA, with a MAPE of 17.5% and an RMSE of 264.2 for WPP 717, although the difference was not statistically significant based on the Diebold-Mariano test. Limitations of this study include the nature of VBD, which represents light detections rather than direct vessel identity or fishing effort, and the still-partial 2026 data. Future research should integrate VBD with AIS, VMS, logbooks, catch data, and oceanographic variables, and should develop machine learning models to improve validation and forecasting capability. Overall, VIIRS DNB has the potential to serve as a complementary source of spatio-temporal information to support WPPNRI-based fisheries monitoring and management.

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